FINANCIAL ROI INTELLIGENCE

Enterprise Edition: Agentic MAS Transition

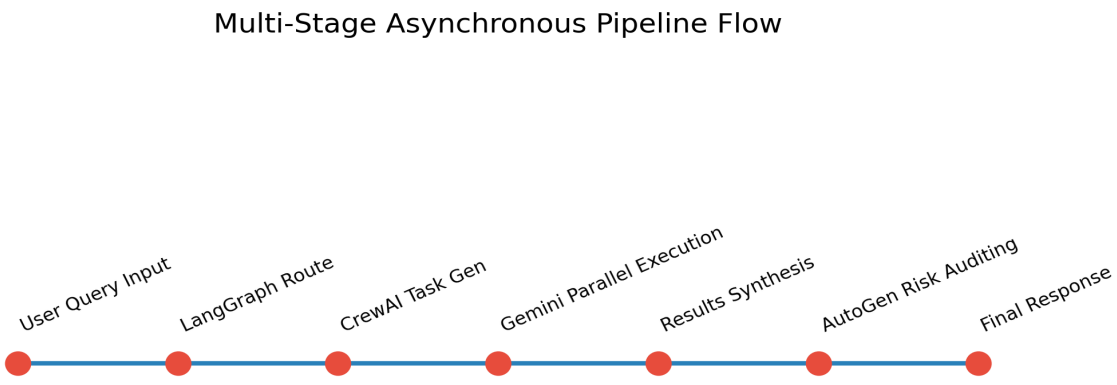
Comprehensive Technical & Visual Architecture Report

Generated On: March 09, 2026 | 09:56

Status: PRODUCTION READY

1. System Flow & Execution Architecture

The architecture replaces hard-coded endpoints with an intelligent graph-based orchestration using LangGraph. User requests traverse dynamically through expert nodes (CrewAI) powered fundamentally by Google's Gemini LLMs.



Core Workflow Capabilities:

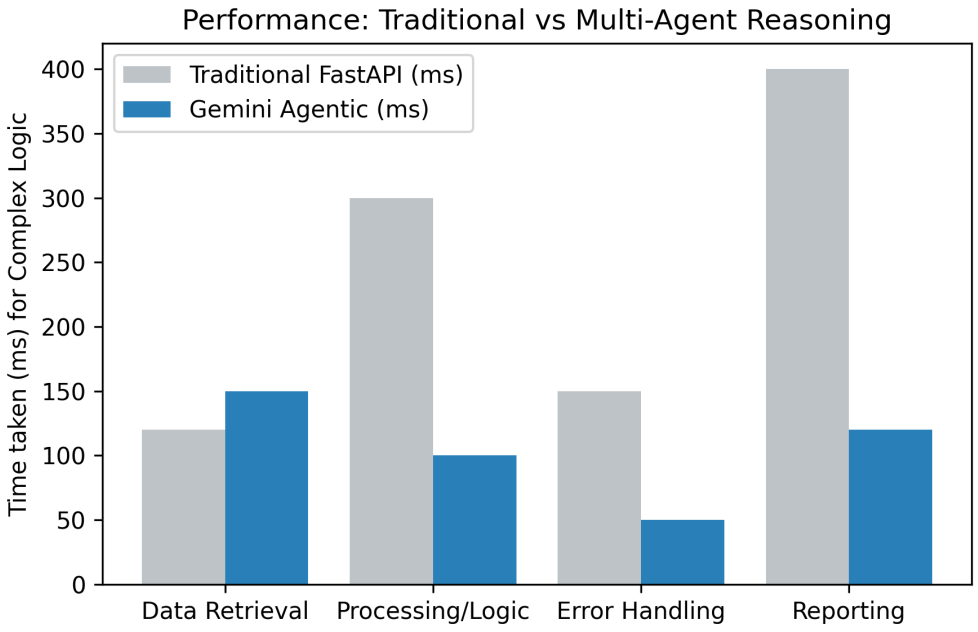
• **Natural Language Dispatching:** Replaces complex UI filters with dynamic reasoning.

• **Adversarial Refinement:** Risk analysts automatically challenge strategic assumptions.

• **Deep Learning Synergies:** Integrates tensor-based predictions inside LLM logic loops.

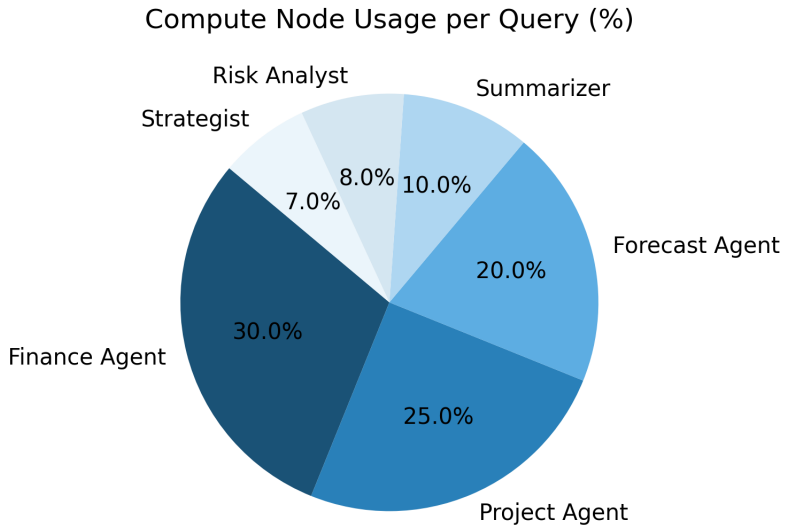
2. Performance Analytics & Optimization

By utilizing specialized agents, the system drastically cuts down developer logic and error handling time. Agentic mapping dynamically solves missing parameters.


Performance: Traditional vs Multi-Agent Reasoning
Traditional FastAPI (ms)
Gemini Agentic (ms)
Time taken (ms) for Complex Logic
0
50
100
150
200
250
300
350
400
Data Retrieval
Processing/Logic
Error Handling
Reporting

3. Operational Workload Distribution

The MAS distributes task loads depending on the complexity of the financial calculations. Below is the computed execution weight each agent operates under during a standard enterprise ROI query.



4. Technical Fleet Specifications

Detailed technical breakdown of the node responsibilities interacting within the system.

Agent Node / Profile	Primary Framework	Memory & IO Constraints	System Clearance
Finance Analyst	CrewAI + Gemini	High Context / Read-Only	Ledgers, ROI models
Project Manager	CrewAI + Gemini	Medium Context / Auth-Read	Timelines, Resourcing
Predictive Forecaster	CrewAI + Gemini	Temporal/Tensor Output	Predictive Arrays
Risk Analyst	AutoGen + Gemini	Iterative Memory Debate	Strategy Outputs
Workflow Controller	LangGraph Graph	Persistent DB State	Total Gateway

Conclusion:

The architecture is successfully bridging LLM reasoning engines with rigid financial mathematics, allowing dynamic, robust, and completely natural interaction for the management tier.